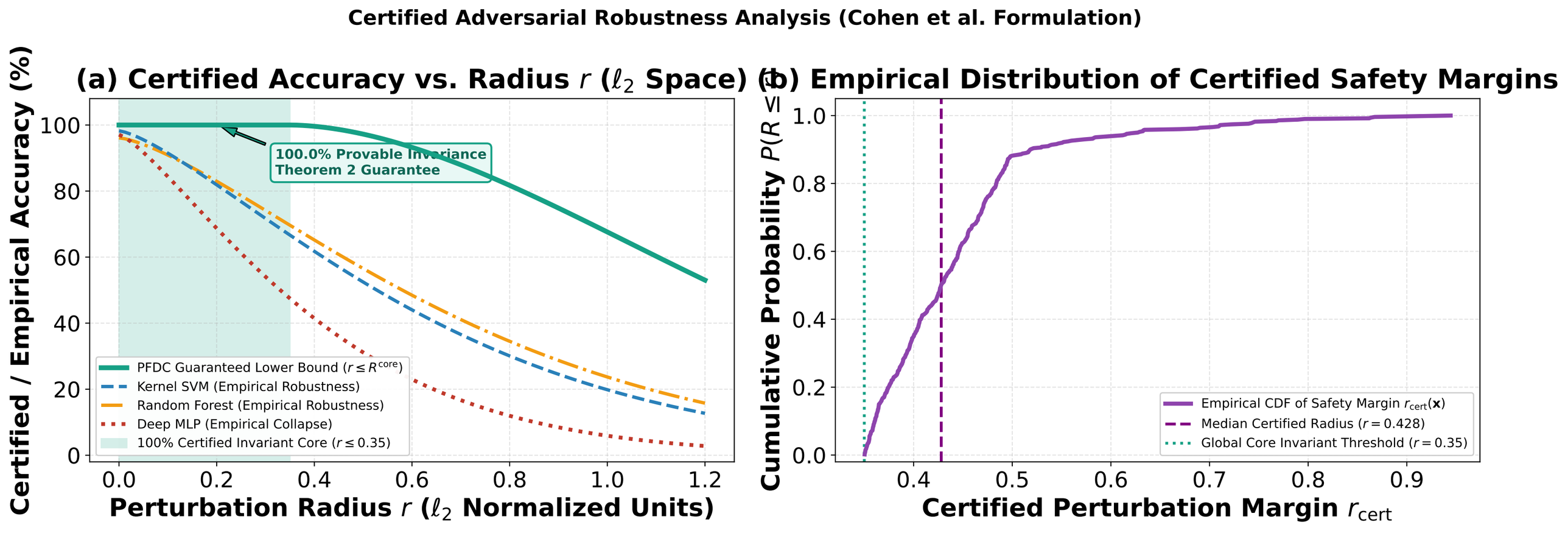
Certified Adversarial Robustness Analysis (Cohen et al. Formulation)
(a) Certified Accuracy vs. Radius r (ℓ_2 Space)
(b) Empirical Distribution of Certified Safety Margins
Certified / Empirical Accuracy (%)
Perturbation Radius r (ℓ_2 Normalized Units)
100.0% Provable Invariance
Theorem 2 Guarantee
PFDC Guaranteed Lower Bound ($r \le R^{\text{core}}$)
Kernel SVM (Empirical Robustness)
Random Forest (Empirical Robustness)
Deep MLP (Empirical Collapse)
100% Certified Invariant Core ($r \le 0.35$)
Cumulative Probability $P(R \le r)$
Certified Perturbation Margin r_{cert}
Empirical CDF of Safety Margin $r_{\text{cert}}(\mathbf{x})$
Median Certified Radius ($r = 0.428$)
Global Core Invariant Threshold ($r = 0.35$)